

Shaker comparison

The four vibration test systems configured in `config/shakers`, generated on 2026-10-06 from the shaker files.

Shakers compared

- **TIRA TV 51110** (- - shaker tv51110): TIRA Vibration Test System TV 51110: exciter S 51110 with power amplifier BDA 120
- **TIRA TV 52110** (- - shaker tv52110): TIRA Vibration Test System TV 52110: exciter S 52110 with power amplifier BDA 120
- **B&K 4809** (- - shaker bk4809): Brüel & Kjær Vibration Exciter Type 4809 with Power Amplifier Type 2718
- **B&K 4801/4812** (- - shaker bk4801_4812): Brüel & Kjær Exciter Body Type 4801 with General Purpose Head Type 4812 and Power Amplifier Type 2707

At a glance

The **B&K 4801/4812** is the only heavy-payload system: about 4.5 times the random force of the TV 51110 and a 13.5 kg static payload limit, but it weighs 80 kg, needs three-phase mains and must run with its blower. The **TIRA TV 51110** and **TV 52110** are complete small systems with the same 100 N sine force; the TV 51110 has more random force (70 vs. 50 N rms) and acceleration, the TV 52110 more stroke (15 vs. 13 mm) and a stiffer suspension. The **B&K 4809** has the lightest moving element and the widest frequency range (to 20 kHz), which suits small, light specimens and high frequencies, but the least force, so payload drops fastest with level.

Ratings

	TIRA TV 51110	TIRA TV 52110	B&K 4809	B&K 4801/4812
Amplifier	TIRA BDA 120 (in system)	TIRA BDA 120 (in system)	B&K 2718	B&K 2707
Frequency range	2 Hz - 7 kHz	2 Hz - 7 kHz	10 Hz - 20 kHz	5 Hz - 10 kHz
Force, sine peak	100 N	100 N	44.5 N	445 N
Force, random rms	70 N	50 N	31.5 N ^d	315 N ^d
Acceleration, sine peak	45 g	50 g	75 g	100 g
Acceleration, random rms	30 g	25 g	53 g ^d	70 g ^d
Stroke (peak-peak)	13 mm	15 mm	8 mm	12.7 mm
Velocity, peak	1.5 m/s	1.5 m/s	1.65 m/s	1.14 m/s
Moving mass	230 g	250 g	60 g	454 g
Suspension stiffness	8 N/mm	13.1 N/mm	12 N/mm	21 N/mm
Armature resonance	6.5 kHz	5.7 kHz	20 kHz	7.2 kHz
Static payload limit	5.3 kg ^d	10 kg ^d	4.9 kg ^d	13.5 kg
Sag per kg (vertical)	1.23 mm	0.75 mm	0.82 mm	0.47 mm
Exciter mass	12 kg	36 kg	8.3 kg	80 kg; 3-phase mains
Cooling	none needed	none needed	optional air (not used in ratings)	blower in body (required)

^d Derived, not in the datasheet: random ratings as sine peak / $\sqrt{2}$; static payload limit as stiffness $\times$ half-stroke / g (the payload whose weight uses up the half-stroke on a vertical shaker).

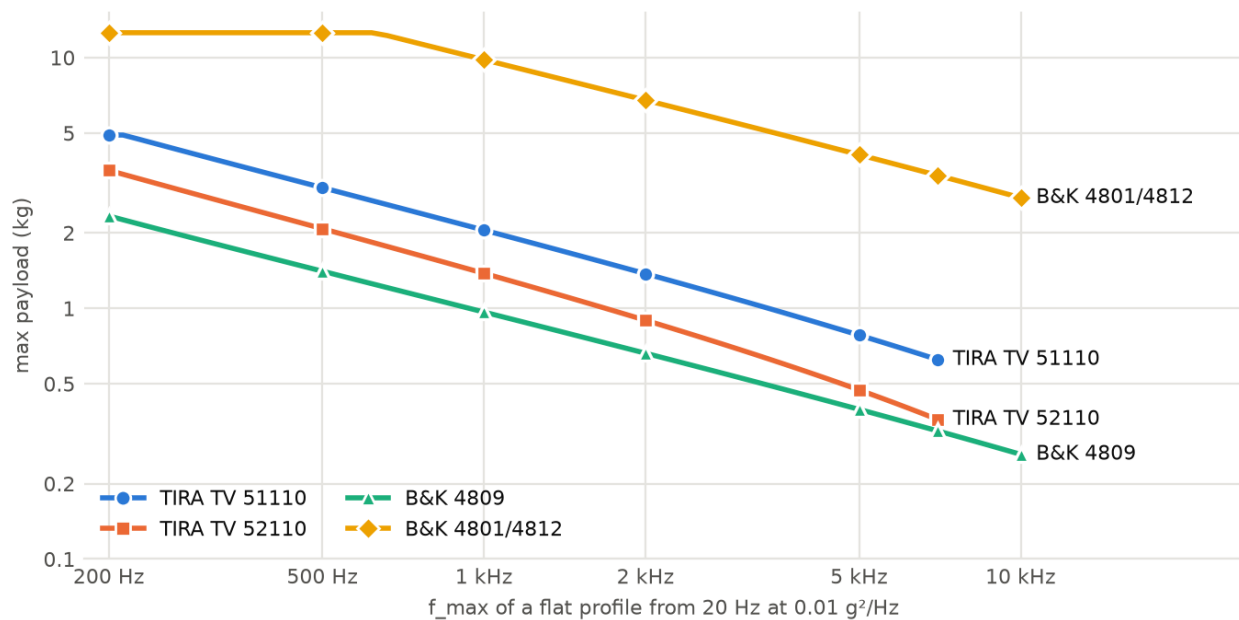
Amplifier and drive limits

	TIRA TV 51110	TIRA TV 52110	B&K 4809	B&K 4801/4812
Amplifier output	120 VA; 22 V rms, 5.5 A rms	120 VA; 22 V rms, 5.5 A rms	75 VA into 3 Ω ; current limit 1-5 A rms	220 VA into 0.5 Ω ; 10 V rms, 22 A rms
Mode and gain	voltage mode only	voltage mode only	voltage, 20 dB gain (assumed)	current, 14 A/V (assumed)
Input for full output	2.47 V rms	2.47 V rms	1 V rms	1.57 V rms
Drive clip (peak)	3.5 V	3.5 V	2.5 V	3.5 V
Drive abort (rms)	1.5 V	1.5 V	0.8 V	1.25 V
Usable with USB-4431 (± 3.5 V AO)	yes	yes	yes	yes

Drive limits are the controller's limits at the amplifier input, set in each shaker file. Amplifier gain settings marked 'assumed' must match the real knob positions.

Payload limits

Maximum payload (accelerometer + fixture + specimen) for flat random profiles from 20 Hz to $f_{\max}$ at $0.01 \text{ g}^2/\text{Hz}$, mounted vertically. Wider bands need more rms acceleration and therefore more force; the 4801/4812 curve is flat at low $f_{\max}$ because its stroke, not its force, limits the payload there.



Flat $0.001 \text{ g}^2/\text{Hz}$: stroke 0.30 mm peak-peak (3σ)

$f_{\max}$ (20 Hz start)	TIRA TV 51110	TIRA TV 52110	B&K 4809	B&K 4801/4812
200 Hz (0.4 g rms)	5.18 kg†	9.82 kg†	4.71 kg†	13.27 kg†
500 Hz (0.7 g rms)	5.18 kg†	7.11 kg	4.58 kg	13.27 kg†
1 kHz (1.0 g rms)	5.18 kg†	4.90 kg	3.18 kg	13.27 kg†
2 kHz (1.4 g rms)	4.84 kg	3.37 kg	2.22 kg	13.27 kg†
5 kHz (2.2 g rms)	2.97 kg	2.03 kg	1.38 kg	13.27 kg†
7 kHz (2.6 g rms)	2.47 kg	1.68 kg	1.16 kg	11.70 kg
10 kHz (3.2 g rms)	-	-	0.96 kg	9.71 kg

Flat 0.01 g²/Hz: stroke 0.96 mm peak-peak (3σ)

f_max (20 Hz start)	TIRA TV 51110	TIRA TV 52110	B&K 4809	B&K 4801/4812
200 Hz (1.3 g rms)	4.91 kg†	3.55 kg	2.33 kg	12.57 kg†
500 Hz (2.2 g rms)	3.03 kg	2.08 kg	1.41 kg	12.57 kg†
1 kHz (3.1 g rms)	2.05 kg	1.38 kg	0.97 kg	9.81 kg
2 kHz (4.4 g rms)	1.37 kg	0.90 kg	0.66 kg	6.76 kg
5 kHz (7.1 g rms)	0.78 kg	0.47 kg	0.40 kg	4.10 kg
7 kHz (8.4 g rms)	0.62 kg	0.36 kg	0.32 kg	3.39 kg
10 kHz (10.0 g rms)	-	-	0.26 kg	2.76 kg

Flat 0.05 g²/Hz: stroke 2.15 mm peak-peak (3σ)

f_max (20 Hz start)	TIRA TV 51110	TIRA TV 52110	B&K 4809	B&K 4801/4812
200 Hz (3.0 g rms)	2.15 kg	1.45 kg	1.01 kg	10.25 kg
500 Hz (4.9 g rms)	1.23 kg	0.79 kg	0.60 kg	6.10 kg
1 kHz (7.0 g rms)	0.79 kg	0.48 kg	0.40 kg	4.13 kg
2 kHz (9.9 g rms)	0.49 kg	0.26 kg	0.26 kg	2.77 kg
5 kHz (15.8 g rms)	0.22 kg	0.07 kg	0.14 kg	1.58 kg
7 kHz (18.7 g rms)	0.15 kg	0.02 kg	0.11 kg	1.27 kg
10 kHz (22.3 g rms)	-	-	0.08 kg	0.98 kg

Unmarked values are limited by force: payload $\leq F_{\text{random rms}} / (a_{\text{rms}} \times g)$ - moving mass. † Limited by stroke on a vertical shaker: static sag plus half the vibration stroke must fit in the half-stroke; mounted horizontally these values rise to the force limit. The stroke barely depends on f_max but grows steeply for a lower start frequency (about 2.8× at 10 Hz). A heavy payload also lowers the armature resonance, which this force check does not cover.

Notes and caveats

- The B&K values come from old instruction manuals and a product datasheet; the manuals give no random ratings, so those are derived from the sine ratings.
- Amplifier gains for the B&K systems are assumptions (2718 at 20 dB, 2707 in current mode with the gain fully clockwise). Different settings change the input for full output and the drive limits.
- The 4809 can technically run on the 2707, but the 2707's slow rms current trip (50 s or 2.5 s averaging) protects the 4809 coil much less well than the 2718's instantaneous current limiter; keep the 4809 on the 2718.
- Only the simulator and the pre-flight check have been run for the B&K setups; start real tests at a low level (for example --level -12).